Branch/Class/Batch: AI&DS/T.E/A1
Student Roll no.: A16
Student Name: Piyush Haresh Koli

EXPERIMENT NUMBER: 1

AIM: To study different networking commands and network cables & connectors.

OBJECTIVES

1. To understand the purpose of various networking commands.
2. To study different types of network cables and connectors.
3. To identify the applications of networking commands in troubleshooting.
4. To understand the structure and use of network connectors.
5. To verify network connectivity using command-line utilities.

THEORY:

Networking Commands

Networking commands are command-line utilities used to configure, test, troubleshoot, and monitor network communication between computers. These commands help network administrators identify connectivity issues, check IP configurations, verify communication with remote devices, and analyze network performance.

Common Networking Commands

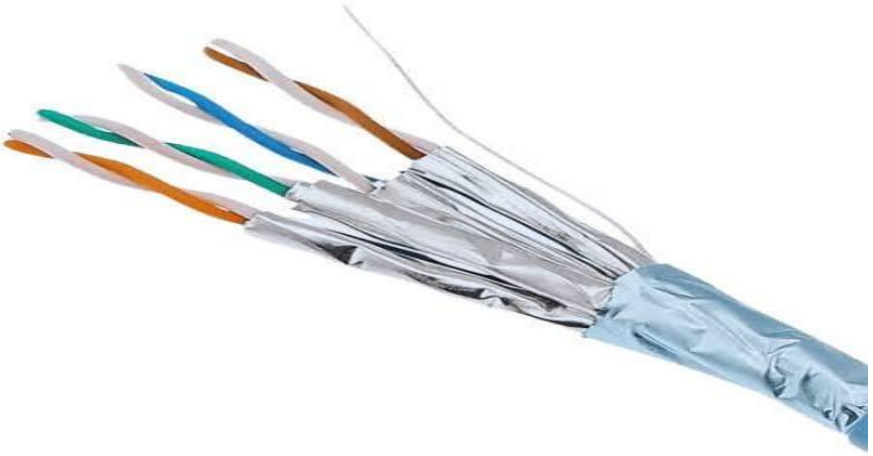
Command	Purpose
ipconfig	Displays the IP address, subnet mask, default gateway, and other TCP/IP configuration details.
ping	Tests connectivity between two devices by sending ICMP Echo Request packets.
tracert	Displays the route taken by packets to reach the destination host.
netstat	Shows active network connections, listening ports, and protocol statistics.
nslookup	Finds the IP address corresponding to a domain name using DNS.
hostname	Displays the name of the local computer.
getmac	Displays the MAC address of the network adapter.
arp -a	Displays the Address Resolution Protocol (ARP) cache.

Branch/Class/Batch: AI&DS/T.E/A1
Student Roll no.: A16
Student Name: Piyush Haresh Koli

Types of Network Cables

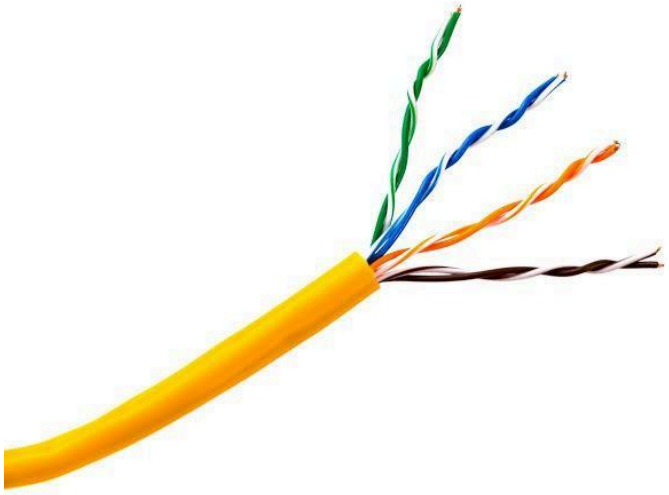
1. Shielded Twisted Pair (STP) Cable

Shielded Twisted Pair (STP) cable consists of pairs of insulated copper wires covered with a metallic shield. The shield protects the cable from electromagnetic interference (EMI) and crosstalk, making it suitable for industrial environments and high-speed networks. STP provides better signal quality than UTP but is more expensive and difficult to install.



2. Unshielded Twisted Pair (UTP) Cable

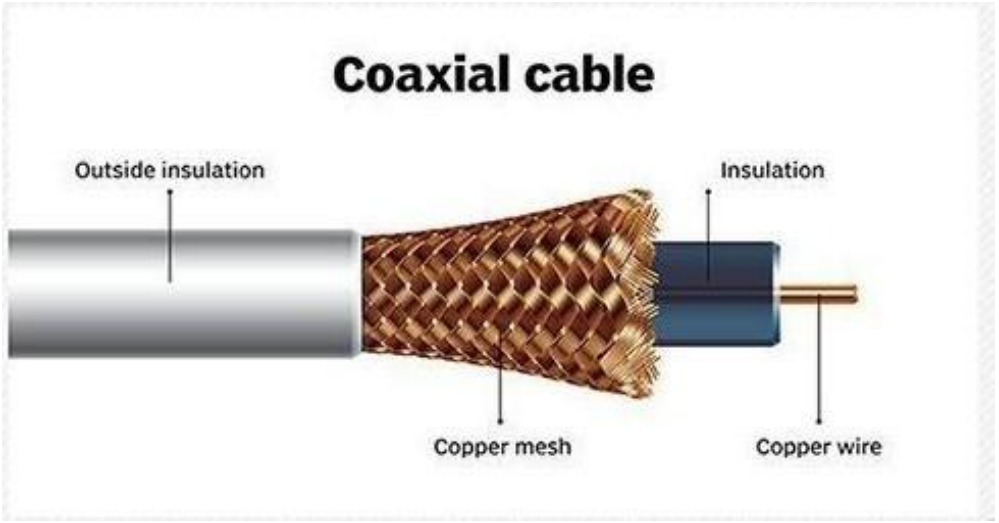
Unshielded Twisted Pair (UTP) cable is the most commonly used network cable in Local Area Networks (LANs). It consists of four pairs of twisted copper wires without any additional shielding. The twisted design reduces interference and supports high-speed data transmission. UTP cables use RJ-45 connectors and are cost-effective, lightweight, and easy to install.



Branch/Class/Batch: AI&DS/T.E/A1
Student Roll no.: A16
Student Name: Piyush Haresh Koli

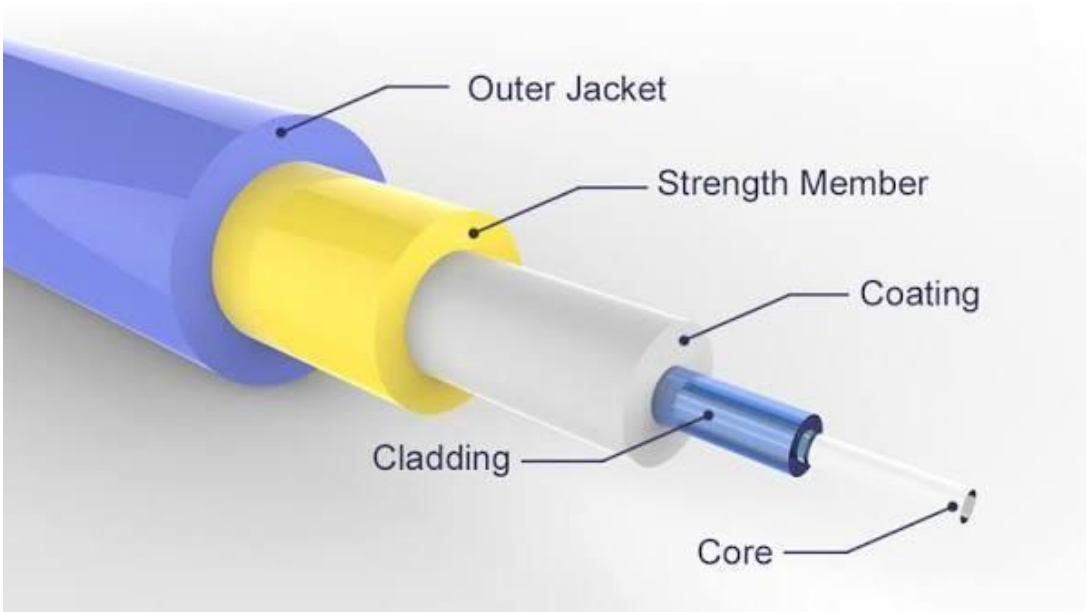
3. Coaxial Cable

A Coaxial cable contains a central copper conductor surrounded by insulation, a metallic shield, and an outer protective jacket. It provides better protection against signal interference than twisted pair cables. Coaxial cables are commonly used in cable television (CATV), broadband internet, CCTV systems, and radio frequency communication. They are durable and suitable for medium-distance communication.



4. Fiber Optic Cable

Fiber Optic cable transmits data in the form of light signals through thin strands of glass or plastic fibers. It offers extremely high bandwidth, very fast data transmission, and supports long-distance communication with minimal signal loss. Fiber optic cables are immune to
electromagnetic interference and are widely used in internet backbones, telecommunications, and data centers.

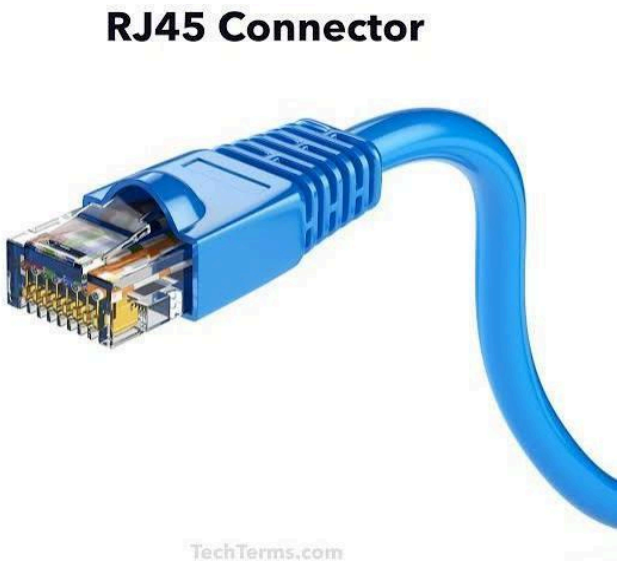


Branch/Class/Batch: AI&DS/T.E/A1
Student Roll no.: A16
Student Name: Piyush Haresh Koli

Types of Network Connectors

1. RJ-45 (Registered Jack-45)

RJ-45 is an 8-pin connector commonly used with UTP and STP Ethernet cables. It is the standard connector for wired Local Area Networks (LANs). RJ-45 supports high-speed data transmission and is used to connect computers, switches, routers, and other networking devices.



2. RJ-11 (Registered Jack-11)

RJ-11 is a small 6-pin connector mainly used in telephone and DSL internet connections. It connects telephones, fax machines, and modems to telephone lines. RJ-11 is smaller than RJ-45 and supports voice communication rather than high-speed networking.



Branch/Class/Batch: AI&DS/T.E/A1
Student Roll no.: A16
Student Name: Piyush Haresh Koli

3. BNC (Bayonet Neill–Concelman) Connector

The BNC connector is used with coaxial cables for secure and reliable signal transmission. It uses a bayonet locking mechanism that allows quick connection and disconnection. BNC connectors are commonly found in CCTV systems, cable television, radio communication, and laboratory equipment.



4. ST (Straight Tip) Connector

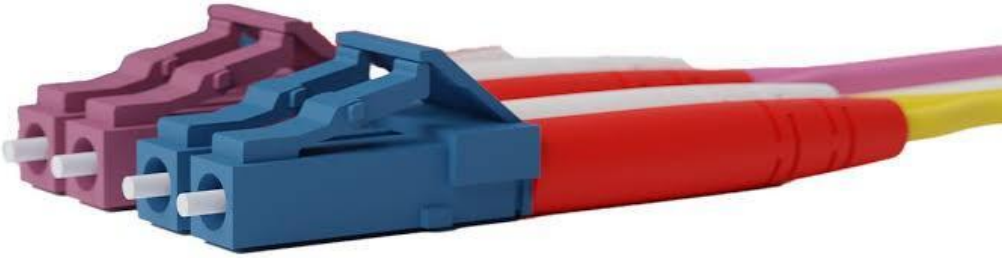
The ST connector is a fiber optic connector that uses a bayonet-style locking mechanism. It is easy to install and provides a secure connection for fiber optic cables. ST connectors are widely used in campus networks, industrial communication systems, and older fiber optic installations.



Branch/Class/Batch: AI&DS/T.E/A1
Student Roll no.: A16
Student Name: Piyush Haresh Koli

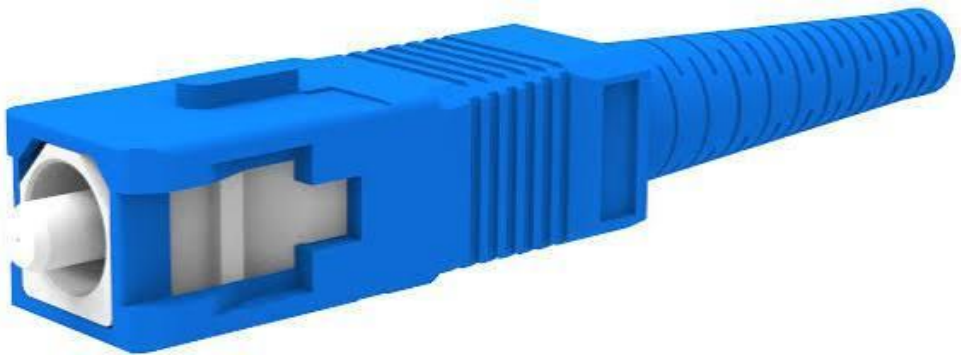
5. LC (Lucent Connector)

The LC connector is a compact fiber optic connector with a push-pull locking mechanism. Due to its small size, it is widely used in high-density networking equipment such as switches, routers, and data centers. LC connectors support both single-mode and multi-mode fiber cables and provide excellent performance.



6. SC (Subscriber Connector)

The SC connector is a square-shaped fiber optic connector that uses a push-pull locking mechanism. It is simple to connect and disconnect while providing low signal loss and high reliability. SC connectors are commonly used in fiber optic communication systems, FTTH (Fiber to the Home), and telecommunications networks.



Branch/Class/Batch: AI&DS/T.E/A1
Student Roll no.: A16
Student Name: Piyush Haresh Koli

DIAGRAM:

```
C:\Users\STUDENT>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::1900:9ee9:e126:5589%4
   IPv4 Address. . . . . . . . . . . : 192.168.3.151
   Subnet Mask . . . . . . . . . . . : 255.255.248.0
   Default Gateway . . . . . . . . . : 192.168.1.254
```

```
C:\Users\STUDENT>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : DESKTOP-80SII67
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek PCIe GbE Family Controller
   Physical Address. . . . . . . . . : F8-BC-12-5D-B5-BC
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::1900:9ee9:e126:5589%4(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.3.151(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.248.0
   Default Gateway . . . . . . . . . : 192.168.1.254
   DHCPv6 IAID . . . . . . . . . . . : 116964370
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2D-85-FA-6E-F8-BC-12-5D-B5-BC
   DNS Servers . . . . . . . . . . . : 103.14.232.100
                                       8.8.8.8
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

```
C:\Users\STUDENT>ping google.com

Pinging google.com [192.178.174.101] with 32 bytes of data:
Reply from 192.178.174.101: bytes=32 time=52ms TTL=113
Reply from 192.178.174.101: bytes=32 time=46ms TTL=113
Reply from 192.178.174.101: bytes=32 time=59ms TTL=113
Reply from 192.178.174.101: bytes=32 time=55ms TTL=113

Ping statistics for 192.178.174.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 46ms, Maximum = 59ms, Average = 53ms
```

Branch/Class/Batch: AI&DS/T.E/A1
Student Roll no.: A16
Student Name: Piyush Haresh Koli

```
C:\Users\STUDENT>tracert google.com

Tracing route to google.com [192.178.174.101]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  192.168.1.254
  2     2 ms     2 ms     2 ms  ternaengg400706.com [59.164.67.217]
  3     2 ms     3 ms     3 ms  172.29.206.46
  4     2 ms     2 ms     2 ms  121.244.40.162.static-mumbai.vsnl.net.in [121.244.40.162]
  5    43 ms    59 ms    33 ms  172.31.167.58
  6   101 ms     *       32 ms  14.141.123.226.static-Chennai.vsnl.net.in [14.141.123.226]
  7    68 ms     *       59 ms  172.25.138.2
  8    62 ms    64 ms    65 ms  115.112.15.74.static-chennai.vsnl.net.in [115.112.15.74]
  9    80 ms    63 ms    66 ms  192.178.120.187
 10    68 ms    74 ms    78 ms  142.251.50.58
 11    72 ms    72 ms    65 ms  142.251.252.95
 12   101 ms     *       69 ms  192.178.254.208
 13    27 ms    26 ms    28 ms  108.170.250.147
 14    30 ms    26 ms    31 ms  142.251.251.186
 15     *        *        *     Request timed out.
 16     *        *        *     Request timed out.
 17     *        *        *     Request timed out.
 18     *        *        *     Request timed out.
 19     *        *        *     Request timed out.
 20     *        *        *     Request timed out.
 21     *        *        *     Request timed out.
 22    64 ms    67 ms   102 ms  lcbomo-in-f101.1e100.net [192.178.174.101]

Trace complete.
```

```
C:\Users\STUDENT>nslookup google.com
DNS request timed out.
    timeout was 2 seconds.
Server:  UnKnown
Address:  103.14.232.100

*** UnKnown can't find google.com: Query refused
```

```
C:\Users\STUDENT>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    192.168.3.151:49779    dns:https              TIME_WAIT
  TCP    192.168.3.151:49784    216.239.34.223:https   TIME_WAIT
  TCP    192.168.3.151:49785    216.239.34.223:https   TIME_WAIT
  TCP    192.168.3.151:49789    172.217.112.4:https    TIME_WAIT
  TCP    192.168.3.151:49792    dns:https              TIME_WAIT
  TCP    192.168.3.151:49793    dns:https              TIME_WAIT
  TCP    192.168.3.151:49799    216.239.32.223:https   TIME_WAIT
  TCP    192.168.3.151:49800    216.239.32.223:https   TIME_WAIT
  TCP    192.168.3.151:49801    lcbomo-in-f100:https   TIME_WAIT
  TCP    192.168.3.151:49805    dns:https              TIME_WAIT
  TCP    192.168.3.151:49806    pnbomb-az-in-f3:https  TIME_WAIT
  TCP    192.168.3.151:49807    lcbomp-in-f84:https    TIME_WAIT
  TCP    192.168.3.151:49813    dns:https              TIME_WAIT
  TCP    192.168.3.151:49815    lcbomp-in-f84:https    TIME_WAIT
  TCP    192.168.3.151:49824    lcdels-in-f188:5228    ESTABLISHED
  TCP    192.168.3.151:49862    dns:https              TIME_WAIT
  TCP    192.168.3.151:49863    lcbome-in-f95:https    TIME_WAIT
  TCP    192.168.3.151:49864    dns:https              TIME_WAIT
  TCP    192.168.3.151:49865    dns:https              ESTABLISHED
  TCP    192.168.3.151:49867    pnbomb-az-in-f3:https  TIME_WAIT
  TCP    192.168.3.151:49868    pnbomb-az-in-f3:https  TIME_WAIT
  TCP    192.168.3.151:49871    162.159.142.9:http     TIME_WAIT
  TCP    192.168.3.151:49876    pt-in-f113:https       TIME_WAIT
  TCP    192.168.3.151:49887    dns:https              TIME_WAIT
  TCP    192.168.3.151:49890    dns:https              TIME_WAIT
  TCP    192.168.3.151:49893    20.190.146.36:https    TIME_WAIT
  TCP    192.168.3.151:49895    172.66.2.5:http        TIME_WAIT
  TCP    192.168.3.151:49902    pu-in-f94:http         TIME_WAIT
  TCP    192.168.3.151:49903    a23-222-31-124:http    TIME_WAIT
  TCP    192.168.3.151:49904    199.232.214.172:http   TIME_WAIT
  TCP    192.168.3.151:49907    20.190.146.36:https    ESTABLISHED
  TCP    192.168.3.151:49908    4.213.25.242:https     ESTABLISHED
  TCP    192.168.3.151:49911    20.190.146.36:https    TIME_WAIT
  TCP    192.168.3.151:49914    199.232.214.172:http   TIME_WAIT
  TCP    192.168.3.151:49918    98.70.192.224:https    TIME_WAIT
  TCP    192.168.3.151:49920    74.176.0.66:https      TIME_WAIT
  TCP    192.168.3.151:49922    a23-222-31-124:http    TIME_WAIT
  TCP    192.168.3.151:49923    199.232.214.172:http   TIME_WAIT
```

Branch/Class/Batch: AI&DS/T.E/A1
Student Roll no.: A16
Student Name: Piyush Haresh Koli

```
TCP    192.168.3.151:49922    a23-222-31-124:http    TIME_WAIT
TCP    192.168.3.151:49923    199.232.214.172:http   TIME_WAIT
TCP    192.168.3.151:49926    server-18-66-41-80:http  TIME_WAIT
TCP    192.168.3.151:49932    216.239.34.223:https   TIME_WAIT
TCP    192.168.3.151:49936    pu-in-f193:https       TIME_WAIT
TCP    192.168.3.151:49937    lcbomp-in-f139:https   TIME_WAIT
TCP    192.168.3.151:49938    199.232.210.172:http   TIME_WAIT
TCP    192.168.3.151:49940    4.207.44.73:https      TIME_WAIT
TCP    192.168.3.151:49941    162.159.142.9:http     TIME_WAIT
TCP    192.168.3.151:49944    216.239.34.223:https   TIME_WAIT
TCP    192.168.3.151:49948    85.210.196.11:https    TIME_WAIT
TCP    192.168.3.151:49949    4.230.171.124:https    TIME_WAIT
TCP    192.168.3.151:49950    142.251.156.119:https  TIME_WAIT
TCP    192.168.3.151:49951    fx-in-f139:https       TIME_WAIT
TCP    192.168.3.151:49952    85.210.196.11:https    TIME_WAIT
TCP    192.168.3.151:49953    216.239.36.223:https   TIME_WAIT
TCP    192.168.3.151:49954    fx-in-f94:https        TIME_WAIT
TCP    192.168.3.151:49955    216.239.34.157:https   TIME_WAIT
TCP    192.168.3.151:49956    dns:https              TIME_WAIT
TCP    192.168.3.151:49957    pt-in-f95:https        TIME_WAIT
TCP    192.168.3.151:49958    fx-in-f101:https       TIME_WAIT
TCP    192.168.3.151:49959    216.239.34.157:https   TIME_WAIT
TCP    192.168.3.151:49960    dns:https              TIME_WAIT
TCP    192.168.3.151:49961    cm-in-f138:https       TIME_WAIT
TCP    192.168.3.151:49962    lcbome-in-f132:https   TIME_WAIT
TCP    192.168.3.151:49963    lcbomp-in-f93:https    TIME_WAIT
TCP    192.168.3.151:49964    cm-in-f138:https       TIME_WAIT
TCP    192.168.3.151:49965    lcbome-in-f132:https   TIME_WAIT
TCP    192.168.3.151:49966    bom12s10-in-f14:https  TIME_WAIT
TCP    192.168.3.151:49967    sm-in-f119:https       TIME_WAIT
TCP    192.168.3.151:49968    fx-in-f149:https       TIME_WAIT
TCP    192.168.3.151:49969    216.239.38.223:https   TIME_WAIT
TCP    192.168.3.151:49971    whatsapp-cdn-shv-01-bom5:https  TIME_WAIT
TCP    192.168.3.151:49972    whatsapp-cdn-shv-01-bom5:https  TIME_WAIT
TCP    192.168.3.151:49978    whatsapp-cdn-shv-01-bom5:https  TIME_WAIT
TCP    192.168.3.151:49982    216.239.38.223:https   TIME_WAIT
TCP    192.168.3.151:49983    150.171.27.12:https    ESTABLISHED
TCP    192.168.3.151:49984    150.171.27.12:https    ESTABLISHED
TCP    192.168.3.151:49985    162.159.142.9:http     ESTABLISHED
TCP    192.168.3.151:49986    20.190.146.36:https    ESTABLISHED
TCP    192.168.3.151:49987    151.101.195.42:https   ESTABLISHED
TCP    192.168.3.151:49989    150.171.27.12:https    ESTABLISHED
TCP    192.168.3.151:49990    172.215.188.225:https  TIME_WAIT
TCP    192.168.3.151:49992    whatsapp-cdn-shv-01-bom5:5222  ESTABLISHED
```

```
TCP    192.168.3.151:49990    172.215.188.225:https  TIME_WAIT
TCP    192.168.3.151:49992    whatsapp-cdn-shv-01-bom5:5222  ESTABLISHED
TCP    192.168.3.151:49993    151.101.131.4:http     ESTABLISHED
TCP    192.168.3.151:49994    whatsapp-cdn-shv-01-bom5:https  TIME_WAIT
TCP    192.168.3.151:49995    192.168.1.254:8090     TIME_WAIT
TCP    192.168.3.151:49996    lcdels-in-f188:5228    ESTABLISHED
TCP    192.168.3.151:49997    pt-in-f101:https       TIME_WAIT
TCP    192.168.3.151:49998    whatsapp-cdn-shv-01-bom5:https  TIME_WAIT
TCP    192.168.3.151:50004    a-0003:http            ESTABLISHED
TCP    192.168.3.151:50006    whatsapp-cdn-shv-01-bom5:https  TIME_WAIT
TCP    192.168.3.151:50007    pnbomb-az-in-f3:https  TIME_WAIT
TCP    192.168.3.151:50008    a23-64-59-58:http      TIME_WAIT
TCP    192.168.3.151:50011    whatsapp-cdn-shv-01-bom5:https  TIME_WAIT
TCP    192.168.3.151:50015    whatsapp-cdn-shv-01-bom5:https  TIME_WAIT
TCP    192.168.3.151:50016    whatsapp-cdn-shv-01-bom5:https  TIME_WAIT
TCP    192.168.3.151:50017    dns:https              TIME_WAIT
TCP    192.168.3.151:50018    lcbome-in-f102:https   TIME_WAIT
TCP    192.168.3.151:50019    whatsapp-cdn-shv-02-bom5:https  ESTABLISHED
TCP    192.168.3.151:50020    pt-in-f101:https       ESTABLISHED
TCP    192.168.3.151:50021    whatsapp-cdn-shv-03-bom2:https  ESTABLISHED
TCP    192.168.3.151:50022    whatsapp-cdn-shv-03-bom2:https  ESTABLISHED
TCP    192.168.3.151:50023    whatsapp-cdn-shv-01-bom5:https  ESTABLISHED
TCP    192.168.3.151:50024    whatsapp-cdn-shv-01-bom5:https  ESTABLISHED
TCP    192.168.3.151:50025    dns:https              ESTABLISHED
TCP    192.168.3.151:50026    a-0003:http            ESTABLISHED
TCP    192.168.3.151:50027    142.251.156.119:https  LAST_ACK
TCP    192.168.3.151:50028    216.239.32.223:https   ESTABLISHED
TCP    192.168.3.151:50029    whatsapp-cdn-shv-03-bom2:https  ESTABLISHED
TCP    192.168.3.151:50030    whatsapp-cdn-shv-02-bom5:https  ESTABLISHED
TCP    192.168.3.151:50031    a104-91-59-97:https    ESTABLISHED
TCP    192.168.3.151:50032    20.76.201.171:https    ESTABLISHED
TCP    192.168.3.151:50034    whatsapp-cdn-shv-01-bom5:https  ESTABLISHED
TCP    192.168.3.151:50035    a104-120-85-242:https  ESTABLISHED
TCP    192.168.3.151:64376    192.168.1.254:8090     ESTABLISHED
```

Branch/Class/Batch: AI&DS/T.E/A1
Student Roll no.: A16
Student Name: Piyush Haresh Koli

```
  TCP    192.168.3.151:49990    172.215.188.225:https  TIME_WAIT
  TCP    192.168.3.151:49992    whatsapp-cdn-shv-01-bom5:5222  ESTABLISHED
  TCP    192.168.3.151:49993    151.101.131.4:http     ESTABLISHED
  TCP    192.168.3.151:49994    whatsapp-cdn-shv-01-bom5:https  TIME_WAIT
  TCP    192.168.3.151:49995    192.168.1.254:8090     TIME_WAIT
  TCP    192.168.3.151:49996    lcdels-in-f188:5228    ESTABLISHED
  TCP    192.168.3.151:49997    pt-in-f101:https       TIME_WAIT
  TCP    192.168.3.151:49998    whatsapp-cdn-shv-01-bom5:https  TIME_WAIT
  TCP    192.168.3.151:50004    a-0003:http            ESTABLISHED
  TCP    192.168.3.151:50006    whatsapp-cdn-shv-01-bom5:https  TIME_WAIT
  TCP    192.168.3.151:50007    pnbomb-az-in-f3:https  TIME_WAIT
  TCP    192.168.3.151:50008    a23-64-59-58:http      TIME_WAIT
  TCP    192.168.3.151:50011    whatsapp-cdn-shv-01-bom5:https  TIME_WAIT
  TCP    192.168.3.151:50015    whatsapp-cdn-shv-01-bom5:https  TIME_WAIT
  TCP    192.168.3.151:50016    whatsapp-cdn-shv-01-bom5:https  TIME_WAIT
  TCP    192.168.3.151:50017    dns:https              TIME_WAIT
  TCP    192.168.3.151:50018    lcbome-in-f102:https   TIME_WAIT
  TCP    192.168.3.151:50019    whatsapp-cdn-shv-02-bom5:https  ESTABLISHED
  TCP    192.168.3.151:50020    pt-in-f101:https       ESTABLISHED
  TCP    192.168.3.151:50021    whatsapp-cdn-shv-03-bom2:https  ESTABLISHED
  TCP    192.168.3.151:50022    whatsapp-cdn-shv-03-bom2:https  ESTABLISHED
  TCP    192.168.3.151:50023    whatsapp-cdn-shv-01-bom5:https  ESTABLISHED
  TCP    192.168.3.151:50024    whatsapp-cdn-shv-01-bom5:https  ESTABLISHED
  TCP    192.168.3.151:50025    dns:https              ESTABLISHED
  TCP    192.168.3.151:50026    a-0003:http            ESTABLISHED
  TCP    192.168.3.151:50027    142.251.156.119:https  LAST_ACK
  TCP    192.168.3.151:50028    216.239.32.223:https   ESTABLISHED
  TCP    192.168.3.151:50029    whatsapp-cdn-shv-03-bom2:https  ESTABLISHED
  TCP    192.168.3.151:50030    whatsapp-cdn-shv-02-bom5:https  ESTABLISHED
  TCP    192.168.3.151:50031    a104-91-59-97:https    ESTABLISHED
  TCP    192.168.3.151:50032    20.76.201.171:https    ESTABLISHED
  TCP    192.168.3.151:50034    whatsapp-cdn-shv-01-bom5:https  ESTABLISHED
  TCP    192.168.3.151:50035    a104-120-85-242:https  ESTABLISHED
  TCP    192.168.3.151:64376    192.168.1.254:8090     ESTABLISHED
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Prajwal Ashok Zolage> arp -a

Interface: 10.126.86.254 --- 0x9
  Internet Address      Physical Address      Type
  10.126.86.11          2e-03-a0-d4-31-41     dynamic
  10.126.86.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
PS C:\Users\Prajwal Ashok Zolage>
```

Branch/Class/Batch: AI&DS/T.E/A1
Student Roll no.: A16
Student Name: Piyush Haresh Koli

```
C:\Users\STUDENT>hostname
DESKTOP-80SII67
```

```
C:\Users\STUDENT>getmac

Physical Address    Transport Name
=================== ==========================================================
F8-BC-12-5D-B5-BC   \Device\Tcpip_{0B70CB67-4363-42D6-A7D6-4A93E0487474}
```

```
C:\Users\STUDENT>pathping google.com

Tracing route to google.com [192.178.211.102]
over a maximum of 30 hops:
  0  DESKTOP-80SII67 [192.168.3.151]
  1  192.168.1.254
  2  ternaengg400706.com [59.164.67.217]
  3  172.29.206.46
  4  121.244.40.162.static-mumbai.vsnl.net.in [121.244.40.162]
  5  172.31.167.58
  6     *        *        *
Computing statistics for 125 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           DESKTOP-80SII67 [192.168.3.151]
                                 11/ 100 = 11%   |
  1    9ms     11/ 100 = 11%     0/ 100 =  0%  192.168.1.254
                                 45/ 100 = 45%   |
  2   11ms     56/ 100 = 56%     0/ 100 =  0%  ternaengg400706.com [59.164.67.217]
                                  2/ 100 =  2%   |
  3   ---     100/ 100 =100%    42/ 100 = 42%  172.29.206.46
                                  0/ 100 =  0%   |
  4   10ms     58/ 100 = 58%     0/ 100 =  0%  121.244.40.162.static-mumbai.vsnl.net.in [121.244.40.162]
                                 42/ 100 = 42%   |
  5   ---     100/ 100 =100%     0/ 100 =  0%  172.31.167.58

Trace complete.

C:\Users\STUDENT>
```

RESOURCES REQUIRED:

- Computer/Laptop with Windows Operating System
- Command Prompt (CMD)
- Internet or LAN connection
- UTP Cable
- Coaxial Cable
- Fiber Optic Cable
- RJ-45 Connector
- RJ-11 Connector
- BNC Connector
- SC/LC Fiber Connectors

PROCEDURES:

1. Open Command Prompt.
2. Execute the networking commands one by one.
3. Observe the output of each command.
4. Identify different network cables.
5. Identify different connectors.
6. Note the application of each cable and connector.

Branch/Class/Batch: AI&DS/T.E/A1
Student Roll no.: A16
Student Name: Piyush Haresh Koli

RESULT:

The various networking commands were executed successfully, and their outputs were observed. Different network cables (UTP, Coaxial, and Fiber Optic) and connectors (RJ-45, RJ-11, BNC, SC, and LC) were studied and identified. Their functions and applications in computer networking were understood successfully.

CONCULSION:

The experiment helped in understanding the purpose and practical use of networking commands for diagnosing and troubleshooting network problems. Different types of network cables and connectors were studied, and their applications in modern communication networks were successfully understood.